Literature Review: Robust Metaheuristic Optimization for Algorithmic Trading: A Comparative Study of Optimization Techniques

Introduction

Algorithmic trading has become increasingly popular in recent years due to its ability to automate decision-making using predefined rules and models, enabling faster execution and more systematic trading strategies. Even with the rise of machine learning and deep learning methods, many trading algorithms still rely heavily on technical indicators to generate trading signals and support decision-making. These rule-based strategies are often built using statistical tools such as moving averages and momentum indicators. They remain widely used because they are simple, interpretable, and relatively easy to implement compared to more complex learning-based approaches. Rule-based trading systems are highly sensitive to hyperparameters such as window lengths, threshold values, and stop-loss or take-profit levels. Poorly chosen or inconsistently tuned parameters, along with the excessive use of technical indicators, can lead to back-test overfitting rather than genuine predictive power in the features [1]

Limitations Of Existing Approaches

Existing trading algorithms often struggle with parameter selection and tuning. As financial data becomes increasingly complex featuring with higher dimensionality, nonlinear relationships and multiple objectives where traditional optimisations methods based on mathematical formulation and single-solution approaches insufficient to meet the needs of the complex solution space [2]. These methods rely on an assumption of smooth and differentiable objective function, whereas the optimisation of trading strategies generally occurs within a non-convex and non-differentiable search space, restricting the effectiveness of traditional and gradient-based methods. Developing an effective trading strategy requires more than simply choosing technical indicators; the main challenge lies in optimising these parameters to ensure the strategy remains robust and performs consistently across different market conditions.

Metaheuristic Optimisation

To address these challenges, we employ metaheuristic optimisation techniques to approximate optimal hyperparameter configurations. These methods are inspired by natural processes such as evolution, swarm intelligence, and collective behaviour, enabling efficient exploration of complex, high-dimensional, and non-convex search spaces without requiring gradient information or explicit directional guidance.

In this work, we adopt Differential Evolution (DE)**,** a population-based stochastic optimisation algorithm designed for global optimisation over continuous domains. DE is particularly well-suited for non-differentiable, nonlinear, and multimodal objective functions, which are common in financial trading problems. Its search mechanism is based on vector differentials within a population, allowing robust exploration and exploitation of the parameter space.

Proposed Approach for Solving the Current Problem

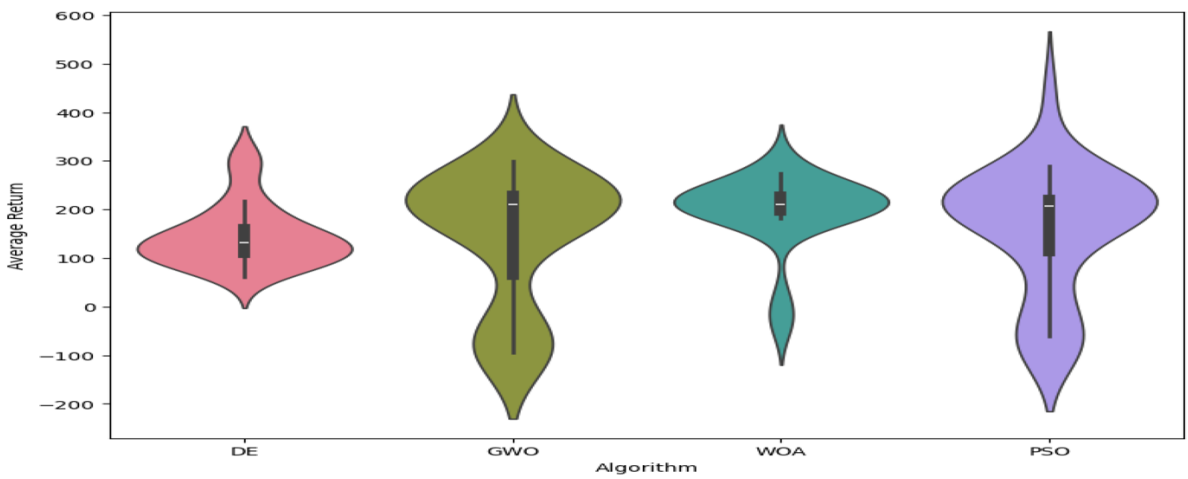
As a parameter optimisation task, where different strategy structures are treated as configurable models within a shared search framework. Each strategy (SMA crossover, MACD, HIGH/LOW, and a generalised moving average model) is defined by a set of

parameters such as window sizes, weights, and smoothing factors. These parameters determine trading signals, which are evaluated through a consistent back-testing process that simulates buy/sell decisions over historical price data. We build on the existing work "Robust Metaheuristic Optimisation for Algorithmic Trading," which employs a static 4-SMA trading model, and adapt it for our study. Instead of focusing on comparing different metaheuristic algorithms, we use this framework to optimise the parameters of the trading strategy and identify the best-performing configuration for improved strategy design.

A population-based optimisation algorithm (Differential Evolution) is used to search this parameter space. Candidate solution represents different parameter configurations, which are iteratively improved based on their trading performance. We use the 4-SMA crossover as the baseline model [4] and extend it to explore progressively more complex trading strategies. This includes 2D and 3D crossover systems, a 4D SMA model, and a 14D generalised framework, where multiple indicators are combined with learnable parameters and find the optimal model his allows both simple and highly generalised models to be optimised under the same framework, highlighting how increasing model flexibility expands the search space while making optimisation more challenging.

Discussion

Based on the Four-SMA crossover results, the Wilcoxon Mann Whitney test indicates that Differential Evolution (DE) achieves higher average annualised returns (107%) with lower variance across runs, suggesting better consistency. However, its performance is sensitive to parameter settings, particularly the mutation factor (F = 0.4 to 0.8) and crossover rate (CR = 0.5 to 0.9).



All metaheuristic methods improve performance over non-optimised strategies when evaluated using rolling-window back testing. The Whale Optimisation Algorithm (WOA) achieves the highest single-run return (120%), while Particle Swarm Optimisation (PSO) shows aggressive behaviour with high win rates but larger drawdowns. Grey Wolf Optimiser (GWO) provides a more balanced performance but with a lower profit factor.

Although direct comparison is influenced by parameter choices, the results suggest that DE offers the most reliable performance, making it a suitable candidate for further experimentation and parameter optimisation.

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